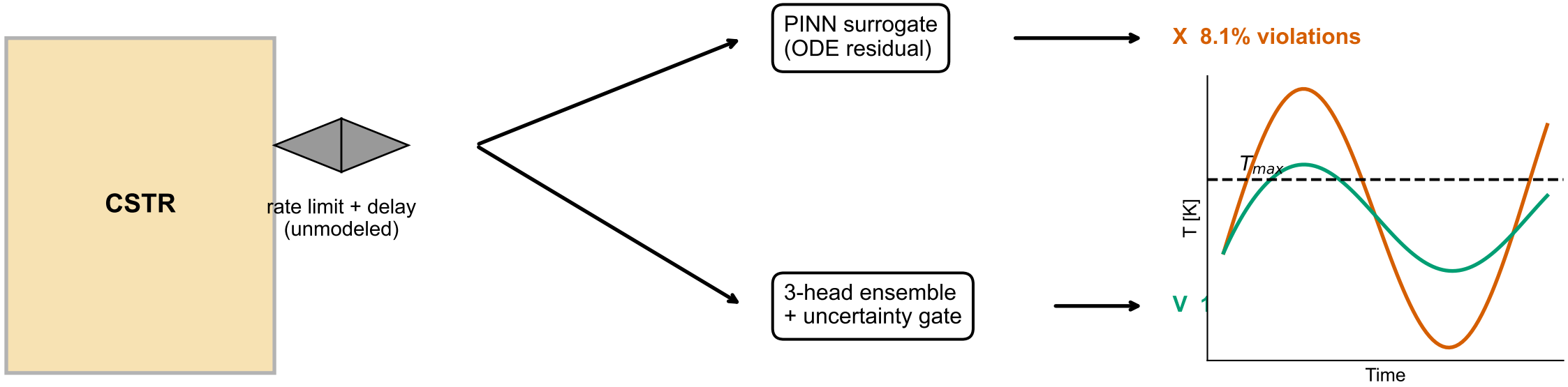
CSTR
rate limit + delay
(unmodeled)
PINN surrogate
(ODE residual)
3-head ensemble
+ uncertainty gate
X 8.1% violations
V
T_{max}
T [K]
Time